
Linking Fatal Crash Data to Safety Ratings

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Does a Car's Safety Rating Truly Impact Fatal Crash Numbers?

Safety ratings are something every car has, and most tests must be passed for it to reach the market. Can a one star safety rating increase the likelihood of a crash being fatal?

From car buyers to insurance agencies, finding the true impact of the ratings could be extremely helpful in predicting when a crash would be fatal.

Gathering Data

National Highway Traffic Safety Administration

Fatal Accident Reporting System Safercar_data.csv

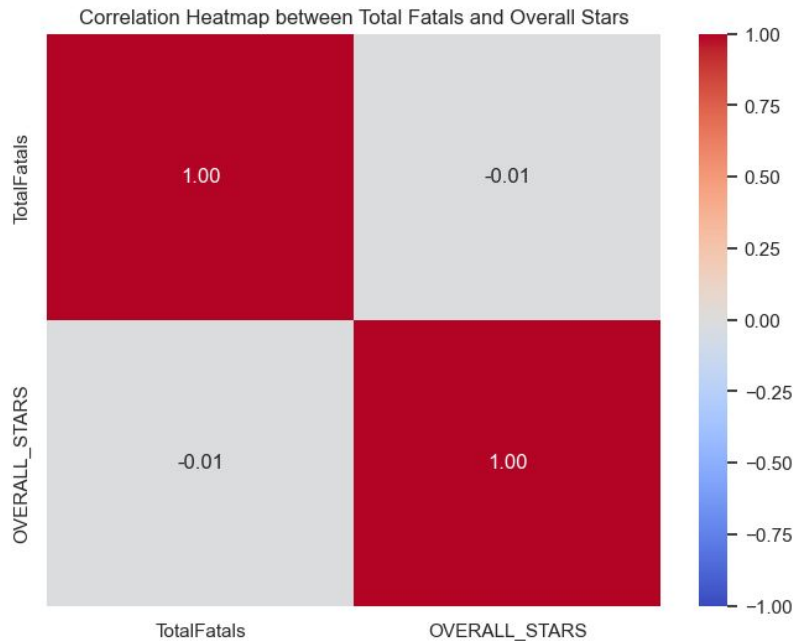
Being a government agency, the NHTSA api is free to use and has all the FARS data from 2010-2021

By using FARS, we only get fatal crashes as part of our data.

Provided by the same administration, this csv has the safety tests and ratings for each vehicle.

Data Pipeline

- Build a Dataframe of Make and Models that exist in the FARS api dataset.
 - Add the number of fatal crashes per year for each model.
 - Add the safety ratings to each car.
 - Clean data of duplicates/outliers.
 - Send to Postgres SQL database.
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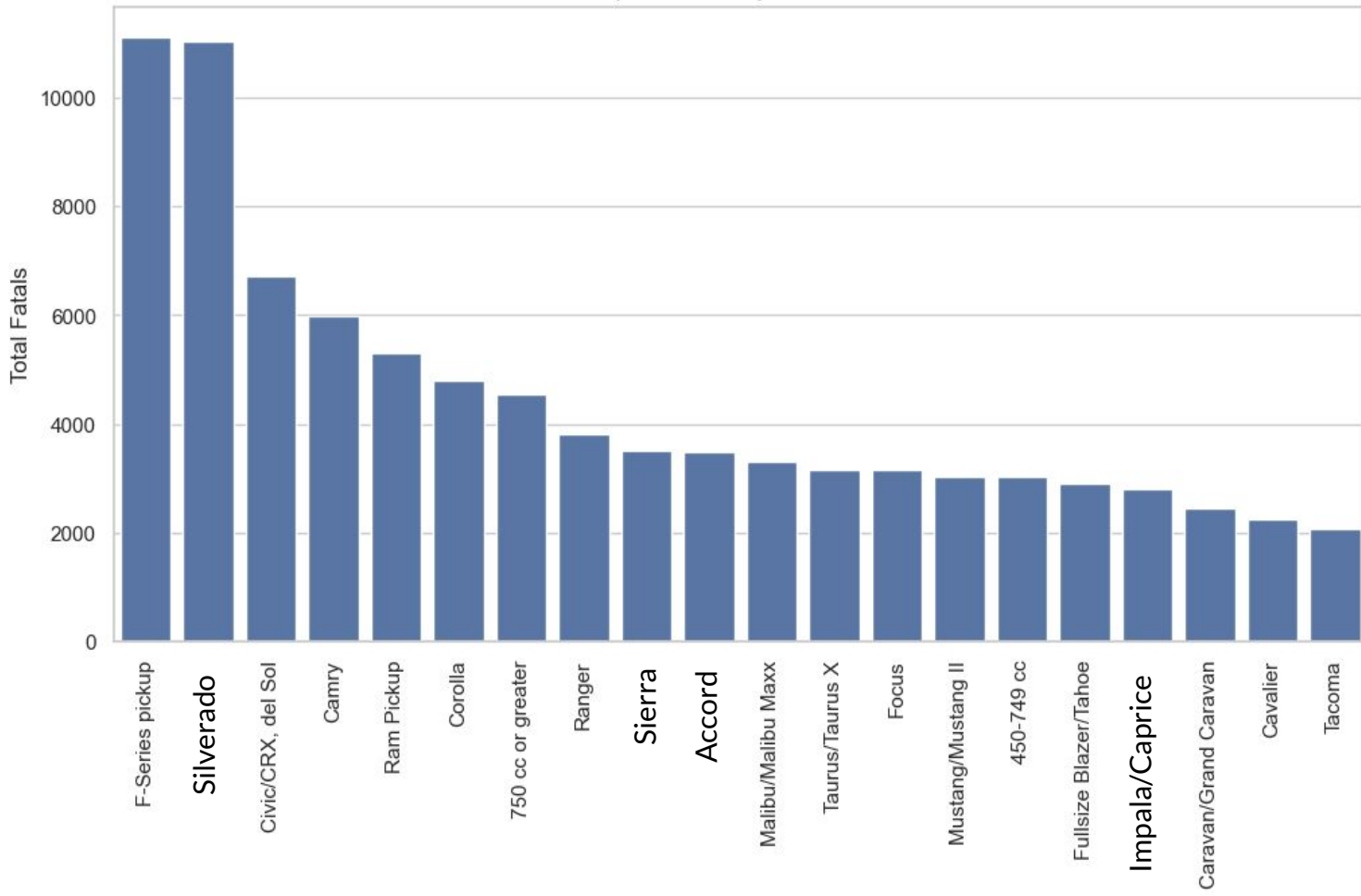


Data Analysis Results & Visualizations

A correlation heatmap gives an extremely strong correlation between Total Fatalities and the car's Overall Star rating. However, it's not entirely what it seems.

With almost all the cars on the road being 5 stars, nearly all fatalities are from 5 star vehicles. This is a perfect example of "Correlation does not equal causation."

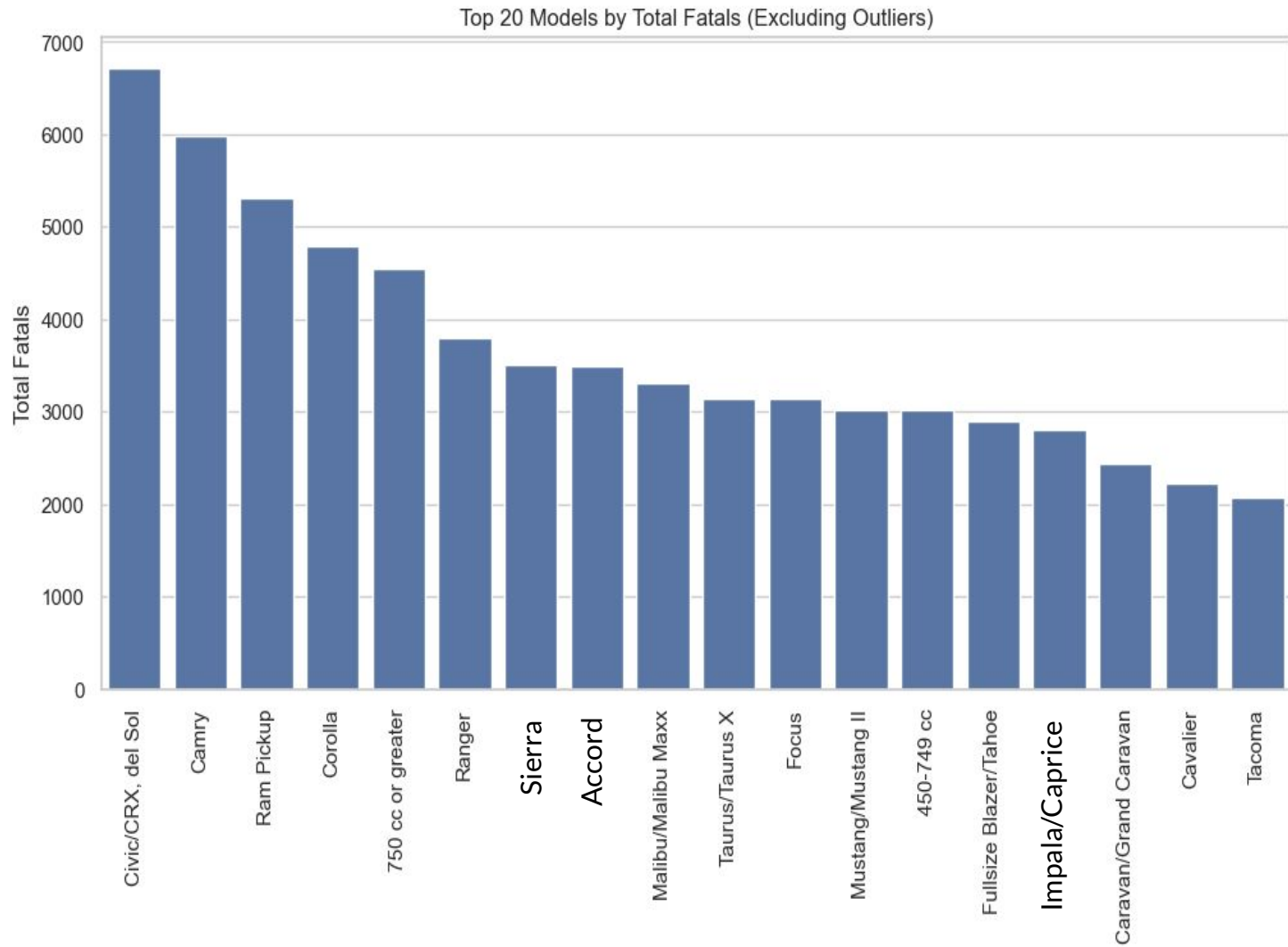
Top 20 Models by Total Fatalis



With a correlation ratio inconclusive, let's investigate the total number of fatalities for each vehicle model.

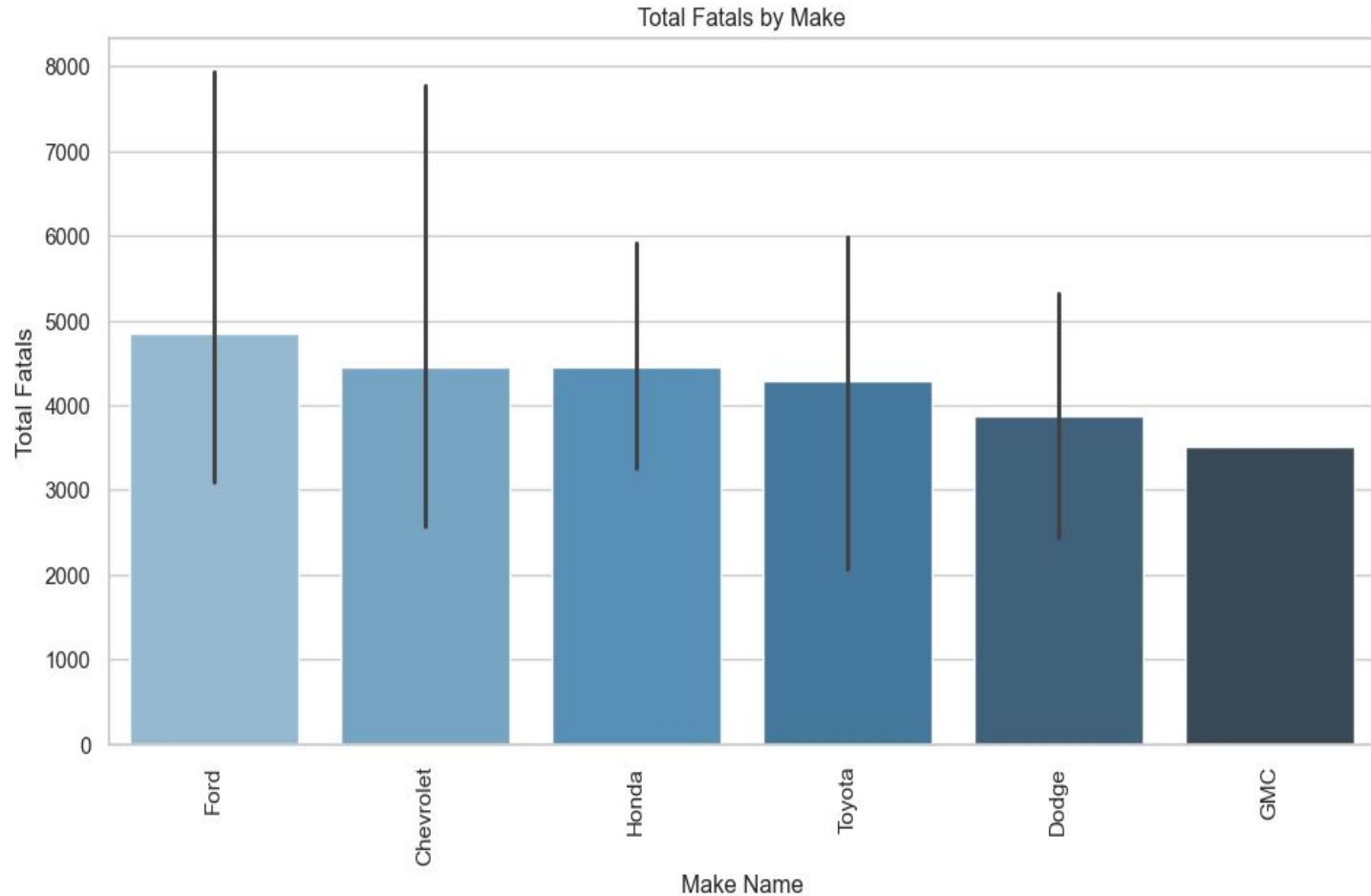
Of the top 20 models, the vast majority of fatalities are from two of the most popular vehicles in the country.

To show the impact of the two most popular cars being removed, here is the graph without them.

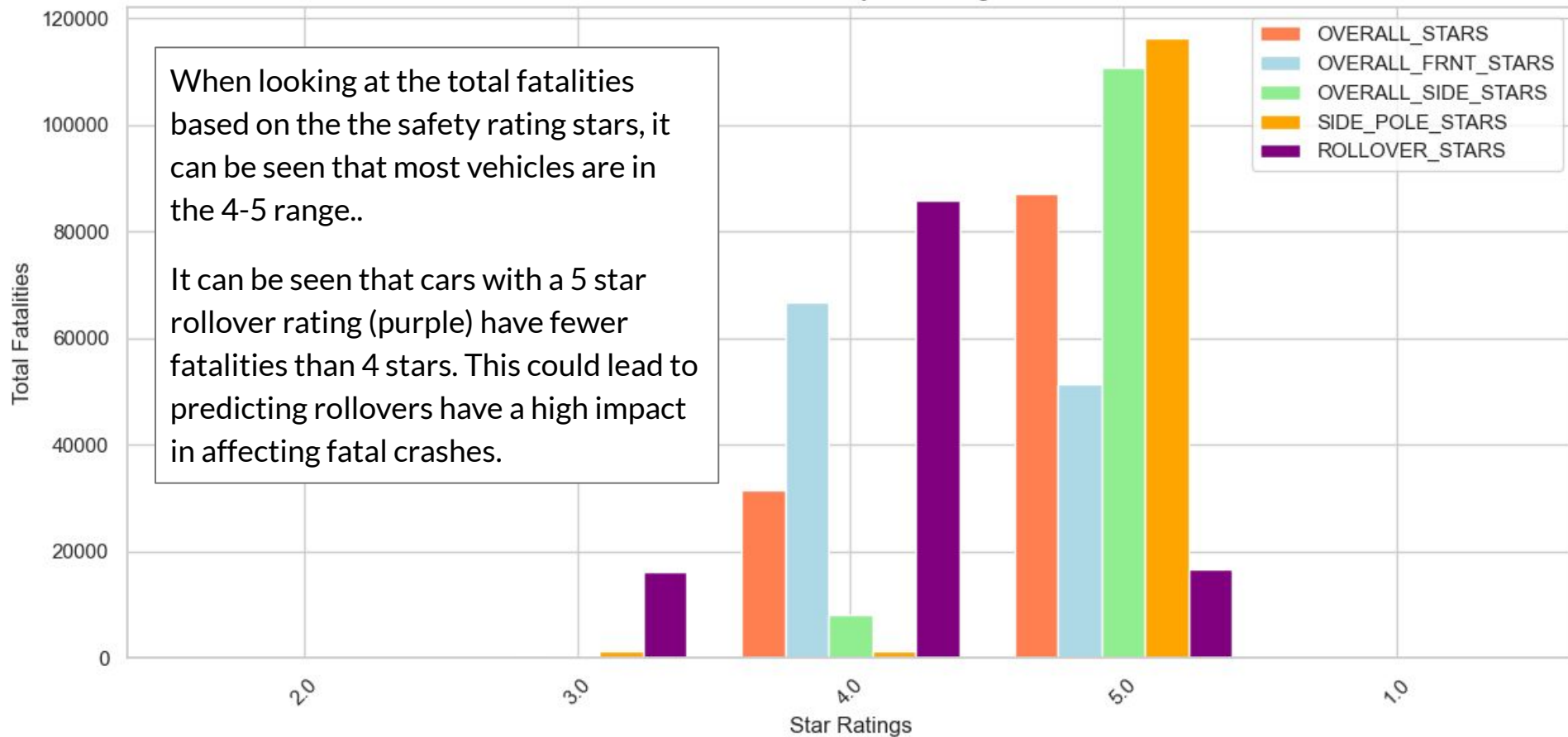


In a broader view, we can look at the total number of fatalities by make (manufacturer).

It is clear that popularity and sheer numbers of vehicles can saturate the data into a skew.



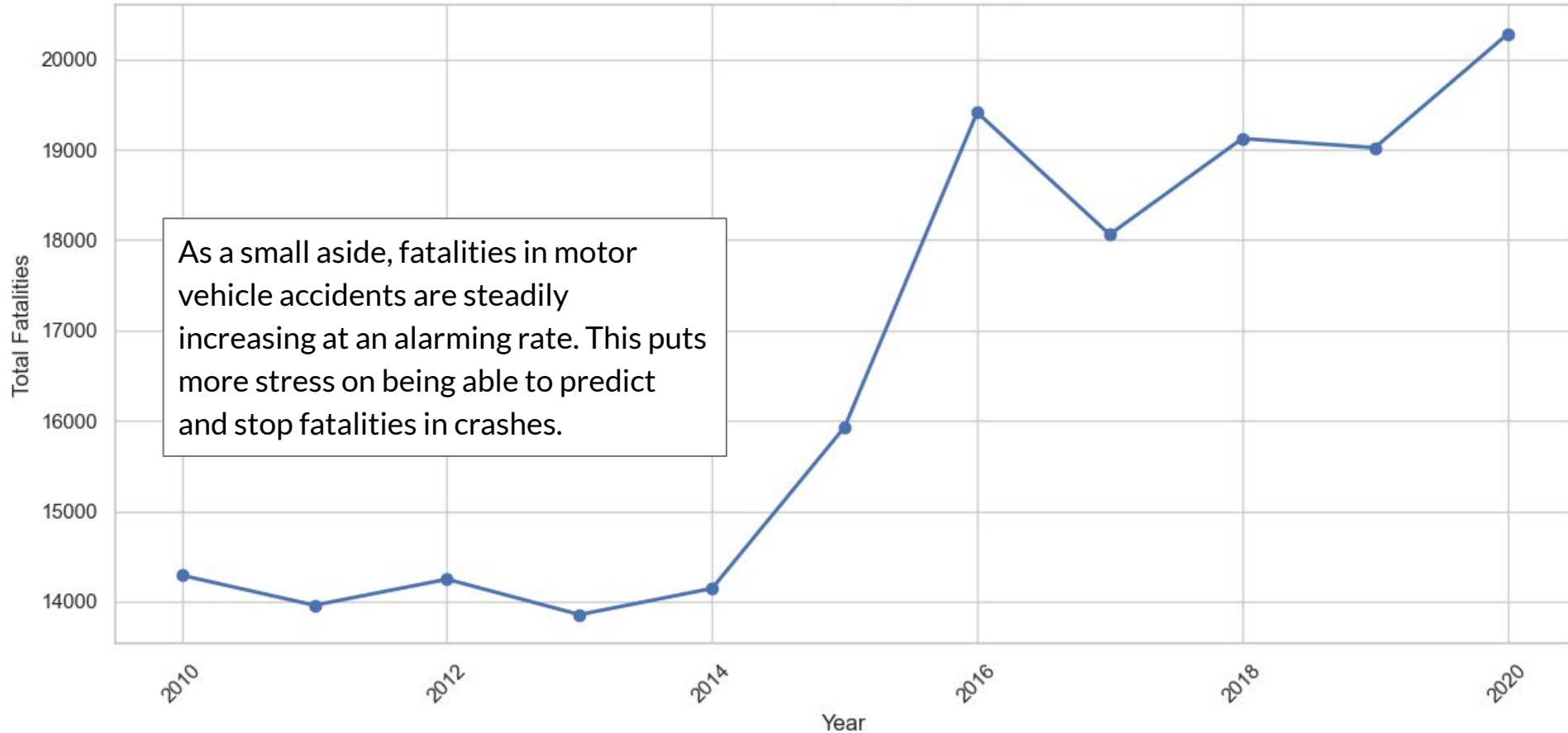
Total Fatalities by Star Ratings



When looking at the total fatalities based on the the safety rating stars, it can be seen that most vehicles are in the 4-5 range..

It can be seen that cars with a 5 star rollover rating (purple) have fewer fatalities than 4 stars. This could lead to predicting rollovers have a high impact in affecting fatal crashes.

Total Fatalities by Year (2010-2020)



Conclusion

Next Steps

While this project wasn't the most groundbreaking discovery, it serves as a resource to point efforts into further refining the causes of fatal crashes for potential prediction. It indicates that more weight should be placed on the type of crash and where the car was damaged.

Overall interpretation

With most cars already having high ratings, and a fatal crash being dependent on so many outside variables, a safety rating alone can't predict a fatal crash. The best data lead is that a better rollover rating can lower the chances of a fatal crash.

**Thank you for
attending our
presentation.**

